

MD Islam (Tamim)

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🔗 <https://tamimcse.github.io/>

EDUCATION

Kent State University

PhD in Computer Science

📅 Aug 2013 – Aug 2022

📍 Kent, OH

BUET

Bachelors in Computer Science and Engineering

📅 Aug 2003 – Dec 2008

📍 Dhaka, Bangladesh

TECHNICAL SKILLS

C++

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RESEARCH AREA

Computer Architecture

Network Algorithms

RESEARCH PROJECTS

C2RTL: A High-Level Synthesis tool

- Developed a high-level synthesis tool named C2RTL that can generate synthesizable Verilog RTL for pipelined ASIC from C code.
- It was designed as a GCC plugin. It takes intermediate code produced by GCC as an input and generates a control and data-flow graph (CDFG) for that. It then performs scheduling and MUX tree generation before producing the Verilog code.
- Evaluated the generated Verilog code with iVerilog and OpenROAD

CP-Trie: A Longest Prefix Match algorithm in Software and ASIC

- Developed several bitmap and Trie based longest prefix match algorithms such as CP-Trie, Poptrie and SAIL for IPv6 routing table lookup.
- Evaluated the algorithms with the routes from real core routers.

NC-TCP: A congestion control in Linux kernel

- Developed several router assisted congestion control such as XCP, RCP and NC-TCP in Linux kernel.
- Evaluated the protocols for a video streaming applications for various network topologies using Mininet and a GStreamer.

SELECTED PUBLICATIONS

- MD Iftakharul Islam, Javed I Khan "C2RTL: A High-level Synthesis System for IP Lookup and Packet Classification." IEEE HPSR, 2021.
- MD Iftakharul Islam, Javed I Khan "CP-Trie: Cumulative Pop-Count based Trie for IPv6 Routing Table Lookup in Software and ASIC." IEEE HPSR, 2021.
- MD Iftakharul Islam, Javed I Khan "A Network-centric TCP for Interactive Video Delivery Networks (VDN)." IEEE ICNP Workshop PVE-SDN, 2017.

WORK EXPERIENCE

Staff Engineer

Samsung SARC/ACL

📅 April 2023 – Present 📍 San Jose, CA

- Working on the functional modeling of the GPU command processor (CP) in C++. It's the control subsystem of Samsung Xclipse GPU which was a AMD RDNA 4 derived architecture.
- Worked on modeling context management, hardware scheduler, pipeline synchronization, performance counter and memory management features
- Worked on the GPU firmwares and functional modeling of the microcontrollers that run the firmwares.
- Developed GPU system tests to add support for new Vulkan API and GPU driver features.
- Worked on performance analysis of gaming benchmark traces

Software Engineer 3

Juniper Networks

📅 Jan 2022–Mar 2023 📍 Sunnyvale, CA

- Worked on packet forwarding software that programs various network algorithms on to Juniper Express ASIC (C++). This system is used on Juniper PTX routers.
- Worked on various network algorithms such as prefix match, range match, TCAM match and so on.

Graduate Teaching Assistant

Kent State University

📅 Sep 2013-Dec 2021 📍 Kent, OH

- Taught Design Patterns and Object Oriented Programming in C++.

Software Engineer Intern

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📅 May 2015-Aug 2015 📍 Columbus, OH

- Developed a call center simulator (C#).

Software Engineer Intern

RightRez

📅 May 2014-Aug 2014 📍 Bloomington, IN

- Developed a flight search algorithm (C#).